

EDUCATION

University Of Warwick

Bachelor of Science in Mathematics

2nd year, Expected Graduation 2027, Current Grade: 1st

Royal Grammar School Newcastle

A-Levels in Mathematics, Further Mathematics, Physics, Computer Science [4 A*s]

2012-2023

SKILLS

Programming & Mathematics

- **Programming** - Python, experience with pytorch, sklearn, pandas, huggingface
- **Machine Learning** - Unsloth, Qwen, API tools, proficient in ML frameworks
- **Mathematics** - Stochastic processes, probability theory, combinatorics, statistical inference, optimisation, Markov Chains, MCMC Methods

PROJECTS AND EXPERIENCE

Hackathon: Testing Monitoring Models with Encoded Transcripts

Project Writeup

- Stress-tested CoT monitorability by using transcripts of malicious activities with different encodings on Claude Haiku 4.5.
- Generated 500 transcripts from Haiku 4.5, with 250 innocent and 250 malicious, then applied different encoding methods.
- Used the proportion of the transcript encoded (in ROT13) as an independent variable to assess what affects the monitoring model's effectiveness.
- We find that Claude Haiku 4.5 performs worse on half-encoded transcripts, compared to completely encoded transcripts.
- We propose that this is the case because Haiku 4.5 can recognise ROT13 encoding and decode accordingly, but this recognition is easier to do when the entire transcript follows the same encoding.
- We observe that the monitoring model's batch size affects how effective the it is, likely because more context means the model has more information to decode the transcript with.

LLM Continuous Probability Inference

github-repo

- Studied the low probability estimation (LPE) problem, which concerns efficiently judging the probability of a rare output from an AI in order to prevent extreme cases in deployment.
- In the context of the LPE problem, applied a new MCMC method from literature called COLD (Energy-based Constrained Text Generation with Langevin Dynamics)
- Adapted the code for COLD to the custom system used by the current methods in the LPE paper.
- Empirically compared the methods to find COLD outperformed all other methods, per sample, bar ITGIS (Independent Token Gradient Importance Sampling), but had a runtime trade off, taking longer to run versus ITGIS.
- Concluded COLD was not the optimal solution, but was worth considering in future projects.

AI Safety Officer

Effective Altruism Warwick

- Hosted a jam site for the 2026 Apart AI Control Hackathon and led the EA team's project.
- Helped organise and run an AI discussion and up-skilling group.
- Led a team looking at red-teaming a monitoring model using encoded chain-of-thought reasoning in large language models.

Neural Network from Scratch

- Created a basic neural network in Python using only the Math and Random modules.
- Used this architecture with Pytorch to train an AI on a names data set to generate new plausible names. Code is in my GitHub.
- Learned about the general structure and logic behind Neural Networks, and compared different activation functions like ReLU vs GELU.